

# Tanvir Mahmud Mahim

Lecturer | [BRAC University](#) | Dhaka, Bangladesh

📍 Room No : 4E-C55, Department of EEE, BSRM SoE, Merul Badda, BRAC University, Dhaka-1212, Bangladesh

🔗 <https://tanvir-mahmud-mahim.github.io> ✉ [tanvir.mahim@bracu.ac.bd](mailto:tanvir.mahim@bracu.ac.bd) ✉ [tanvirmahim14@gmail.com](mailto:tanvirmahim14@gmail.com)

## 🎓 EDUCATION

|          |                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dec 2022 | <b>B.Sc. in Electrical and Electronic Engineering (EEE), BRAC University</b>                                                                                                                                                                                                                                                                                                                         |
| Jan 2019 | <ul style="list-style-type: none"><li>CGPA: <b>3.77/4.00</b>   Average Marks: <b>86.20%</b>   Academic Honors: <b>High Distinction</b></li><li>Certification: <a href="#">World Education Services (WES) Certified</a>, Equivalent to a US Bachelor's Degree</li><li>Supervisor: <a href="#">Dr. A.H.M.A. Rahim</a> [Retired (Dec 2024)]</li><li>Thesis: <a href="#">BRACU-DSpace-2023</a></li></ul> |

## 🎓 HONORS AND AWARDS

**VC's and Dean's List Award** for academic excellence at undergraduate study (2020-2022)  
**University Merit Scholarship** for academic excellence at undergraduate study (2020-2022)

## 💡 RESEARCH INTERESTS

- Quantum Optics and Photonics
- 2D Quantum Materials
- Microelectromechanical Systems (MEMS)
- Wide Bandgap (WBG) Semiconductor Devices
- Electronic Design Automation (EDA)
- Machine Learning (ML)

## 🔬 RESEARCH EXPERIENCE

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Present   | <b>Quantum Optics and Photonics</b>   <a href="#">Dr. A.S.M. Mohsin</a> , <a href="#">BRACU</a> , <a href="#">EEE</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| July 2025 | <ul style="list-style-type: none"><li>Developed a numerical model of a photonic-molecule 4H-SiC soliton-crystal microcomb enabling squeezing extraction beyond 3 dB limit <a href="#">[Code]</a> <a href="#">[under review]</a></li><li>Theoretically and numerically investigated a pick-and-place diamond micromembrane NV quantum sensor for imaging single vortices and quantifying vortex-induced losses in tantalum superconducting circuits <a href="#">[Code]</a> <a href="#">[under review]</a></li><li>Developed closed-loop ML frameworks for quantum photonic instrumentation and photonic neural inference, in collaboration with the <a href="#">University of Memphis, USA</a>: SPARQ: a spiking RL framework for triaging solid-state single-photon emitters <a href="#">[Code]</a> <a href="#">[under review]</a>; Photon-efficient neural inference over optical links <a href="#">[Code]</a> <a href="#">[under review]</a></li><li>Developed learned generative process digital twins for yield-aware inverse design of photonic sensor front-ends <a href="#">[Code]</a> <a href="#">[under review]</a></li></ul> |
| July 2026 | <b>2D Materials and MEMS</b>   <a href="#">Dr. M. Mosaddequr Rahman</a> , <a href="#">BRACU</a> , <a href="#">EEE</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| July 2024 | <ul style="list-style-type: none"><li>Developed a TCAD-based model relating two Raman phonon signatures to fixed edge charge in monolayer semiconductor nanoribbon transistors. <a href="#">[Code]</a> <a href="#">[under review]</a></li><li>Investigated the combination of strain and ferroelectricity in WSe<sub>2</sub> transistors for fast, low-power nonvolatile logic <a href="#">[Code]</a> <a href="#">[under review]</a></li><li>Developed a fabrication-aware, physics-informed adjoint framework with RL for inverse design of CMUTs and photonic sensors. <a href="#">[Code]</a>, <a href="#">[under review]</a>, <a href="#">[IEEE Sensors Journal 2025]</a></li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| June 2026 | <b>WBG based Electronic Devices</b>   <a href="#">Dr. Nadim Chowdhury</a> , <a href="#">BUET</a> , <a href="#">EEE</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| May 2025  | <ul style="list-style-type: none"><li>Developed physics-guided adjoint co-optimization of a novel monolithic E-mode GaN-on-SOI fractional-N PLL with HEMT varactor, <a href="#">[Code]</a> <a href="#">[under review]</a></li><li>Collaborated with <a href="#">GlobalFoundries, Inc. Santa Clara, USA</a> through the development of physics-guided graph RL framework with an adaptive fuzzy reward for transferable analog transistor sizing. <a href="#">[Code]</a> <a href="#">[under review]</a></li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| June 2024 | <b>Dynamic Control</b>   <a href="#">Dr. A.H.M.A. Rahim</a> , <a href="#">BRACU</a> , <a href="#">EEE</a> <a href="#">[Retired (Dec 2024)]</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| May 2023  | <ul style="list-style-type: none"><li>Investigated multi-machine network control with RL-based fuzzy inference <a href="#">[JESTECH (Elsevier) 2026]</a></li><li>Developed an RL-driven fuzzy-attention inference controller for microgrids. <a href="#">[IEEE TFS 2025]</a></li><li>Developed a novel bifacial module <a href="#">[IEEE TENSYP 2023]</a>, <a href="#">[IEEE JPV 2024]</a></li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

- 2026 **T. M. Mahim**, M. M. Rahman and A. S. N. Huda, "Adaptive Takagi-Sugeno-Kang Fuzzy Inference with Reinforcement Learning to Withstand Extreme Low Voltages of DFIG-WT", *Engineering Science and Technology, an International Journal*, 80, 102435 (2026). [\[Paper\]](#)
- 2025 **T. M. Mahim**, M. M. Rahman and A. H. M. A. Rahim, "Hierarchical Inverse Design Framework for Unit-Cell CMUTs With Attentive Gated Recurrent And Fully Connected Dense Layers", *IEEE Sensors Journal*, 25(13), 26104-26116 (2025). [\[Paper\]](#)
- 2025 **T. M. Mahim**, A. H. M. A. Rahim and M. M. Rahman, "Adaptive Fuzzy Attention Inference to Control A Microgrid Under Extreme Fault On Grid Bus", *IEEE Transactions on Fuzzy Systems*, 33(6), 1815-1824 (2025). [\[Paper\]](#)
- 2024 **T. M. Mahim**, A. H. M. A. Rahim and M. M. Rahman, "Weather Responsive Multidimensional Photovoltaic Efficiency Model for Simulation of Custom-Built Bifacial Panel", *IEEE Journal of Photovoltaics*, 14(5), 848-860 (2024). [\[Paper\]](#)
- 2024 **T. M. Mahim**, A. H. M. A. Rahim and M. M. Rahman, "Review of Mono- and Bifacial Photovoltaic Technologies: A Comparative Study," *IEEE Journal of Photovoltaics*, 14(3), 375-396 (2024). [\[Paper\]](#)

## </> TECHNICAL SKILLS

---

- DEVSIM & Synopsys Sentaurus TCAD
- COMSOL Multiphysics
- Ansys HFSS
- Wannierization (Wannier90)
- PyTorch, Scikit-learn, Keras & TensorFlow
- JAX, Optax, Flax
- GDSFactory, Xyce, OpenVAF & ngspice
- Keysight ADS, Cadence Virtuoso
- Density Functional Theory (QUANTUM ESPRESSO)
- Molecular Dynamics (LAMMPS)
- Micromagnetics (MuMax<sup>3</sup>)

## PROFESSIONAL EXPERIENCE

---

- June 2026 **Adjunct Lecturer, Department of EEE, BRACU**
- June 2025 **Research Assistant, Department of EEE, BRACU**
- Present **Research Assistant, Department of EEE, BRACU**
- June 2023 **Consultant, GIXSystems.ai, USA**
- June 2026 **Consultant, GIXSystems.ai, USA**
- Dec 2025 **Teaching Assistant, Department of EEE, BRACU**
- May 2023 **Teaching Assistant, Department of EEE, BRACU**
- Jan 2022 **Teaching Assistant, Department of EEE, BRACU**
- Courses: Solid-State Device, Numerical Methods and Microprocessor & Embedded System
  - Collaborated with Industries and International Universities
  - Provided Assistance to Capstone Projects of Several Undergraduate Groups
  - Technology development of a 100 V GaN-on-Si power HEMT discrete device (tape-out with X-FAB, Erfurt, Germany)
  - Courses: DC & Magnetic Circuits, Energy Conversion, Signals & Systems and Control Systems

## SYNERGISTIC AND LEADERSHIP ACTIVITIES

---

- Present **Student Member | IEEE (Institute of Electrical and Electronics Engineers)**
- Jan 2021 **Student Member | IEEE (Institute of Electrical and Electronics Engineers)**
- Nov 2023 **Aspire Institute | Harvard University, Cambridge, USA**
- Feb 2023 **Aspire Institute | Harvard University, Cambridge, USA**
- Electron Device Society (EDS)
  - Power and Energy Society (PES)
  - Aspire Leadership Program 2023

## REFEREES

---

**Dr. Md. Mosaddequr Rahman**  
Professor and Chairperson, Department of  
EEE, BRAC University, Bangladesh  
[mosaddeq@bracu.ac.bd](mailto:mosaddeq@bracu.ac.bd)

**Dr. Nadim Chowdhury**  
Associate Professor, Department of EEE,  
BUET, Bangladesh  
[nadim@eee.buet.ac.bd](mailto:nadim@eee.buet.ac.bd)

**Dr. Abu S. M. Mohsin**  
Associate Professor, Department of EEE,  
BRAC University, Bangladesh  
[asm.mohsin@bracu.ac.bd](mailto:asm.mohsin@bracu.ac.bd)

**Dr. Navneet Jain**  
Distinguished Member of Technical Staff,  
GlobalFoundries Inc. USA  
[navneet.jain@globalfoundries.com](mailto:navneet.jain@globalfoundries.com)

**Dr. A.S. Nazmul Huda**  
Associate Professor, Department of EEE,  
BRAC University, Bangladesh  
[nazmul.huda@bracu.ac.bd](mailto:nazmul.huda@bracu.ac.bd)